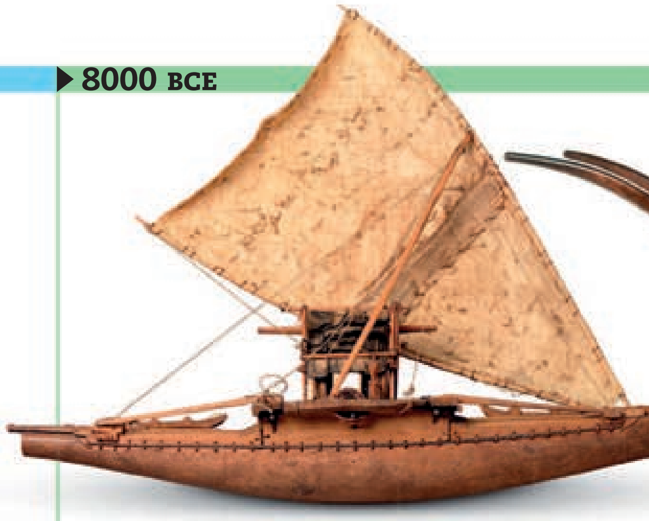
▶ 8000 BCE



MODEL OF FIJIAN BATTLE CANOE

BOAT

Early people needed some form of floating raft to take them fishing and from one island to another. The earliest boats were wooden logs or bamboo trunks tied together, but by around 3000 BCE, people had developed metal tools to cut tree trunks into wooden planks to build the first ships.

▶ 6000 BCE

WOODEN PLOUGH

PLOUGH

People hunted for food until around 8500 BCE when they began to farm the land to grow grains, such as wheat. Wooden ploughs were invented to make use of animal power. Ploughs could be joined to oxen and used to dig up much bigger areas of land.

▶ 3500 BCE

WHEEL

The first wheels were solid wooden discs with a hole through the centre. People needed sharp metal tools to chisel the round shape, which explains why this major invention took a while to arrive. The wheels were connected by a rod called an axle.

WOODEN WHEEL

▶ 900 CE

GUNPOWDER BURNING

GUNPOWDER

Chinese alchemists (early chemists) had been experimenting with chemicals for centuries when a group discovered that a mix of saltpetre, sulphur, and charcoal exploded into flame. The mix was used in fireworks to scare away evil spirits and later in weapons. The recipe was kept from the rest of the world until the 13th century.

◀ 1300

EYE GLASSES

EYE GLASSES

Early peoples such as the Vikings used rock crystals to act as lenses and increase their viewing power. Wearable lenses in the form of eye glasses were invented in the 14th century – probably in Italy, where glassblowing techniques were advanced. These early spectacles were made of two magnifying lenses set into bone, metal, or leather mountings, and were balanced on the nose.

◀ 1590

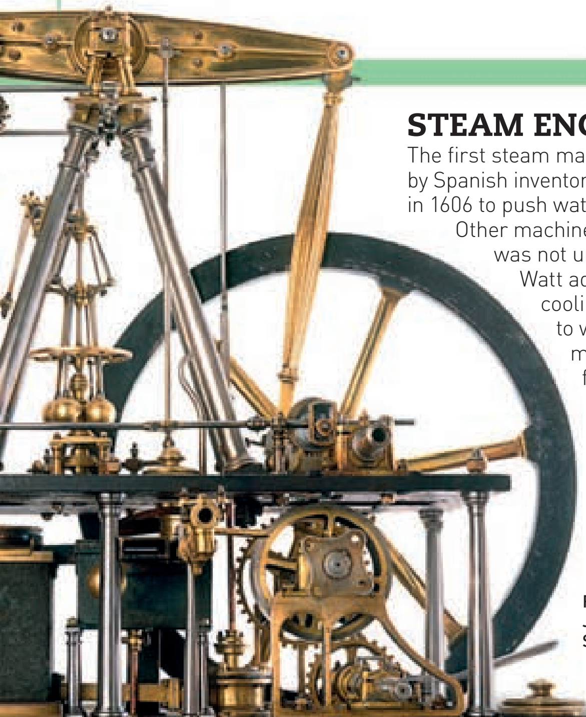
COMPOUND MICROSCOPE

Zacharias Janssen, the son of a spectacles maker in Holland, invented the microscope using a long tube and a mix of curved lenses. In 1665 the Englishman Robert Hooke improved the design and added an oil lamp to light up the specimens. Microscopes have been used by scientists ever since.

REPLICA OF ROBERT HOOKE'S MICROSCOPE

STEAM ENGINE

The first steam machine was designed by Spanish inventor Jerónimo de Ayanz in 1606 to push water out of mines. Other machines followed, but it was not until Scotsman James Watt added a condenser (for cooling the steam back to water) and gears (for making the engine faster) that the steam engine became a useful form of power for factories, mines, farming, and transport.



REPLICA OF JAMES WATT'S STEAM ENGINE

▶ 1928

ANTIBIOTIC PILLS

ANTIBIOTIC

Alexander Fleming's discovery that a mould juice (now known as penicillin) could kill a wide range of bacteria changed the course of modern medicine. Today, there are many types of antibiotics, targeting bacteria, fungi, and parasites.

▶ 1946

COMPUTER

Developed for the US government, the world's first electronic general-purpose computer was called ENIAC: Electronic Numerical Integrator and Computer. This huge computer led the way for smaller and more powerful computers in the decades to come.

COMMODORE (PERSONAL) COMPUTER FROM 1977

▶ 1957

SPACE SATELLITE

The Soviet Union put the first satellite into space on 4 October 1957. Called Sputnik 1, it was the size of a beach ball, and took 98 minutes to orbit Earth. This marked the beginning of the Space Age.

SPUTNIK 1

▶ 1973

1990s MOBILE PHONE

MOBILE PHONE

Martin Cooper, working at Motorola in the USA, developed and demonstrated the first mobile phone. It was the size of a brick and would not be sold to the general public for another ten years, but it marked the start of mobile personal communication systems.

▶ 1989

WORLD WIDE WEB

In the 1970s Vinton Cerf developed a system that allowed mini-networks of computers all over the world to send files to each other. Then in 1991 Tim Berners-Lee introduced a World Wide Web of information that anyone with an online computer could access, and helped to create the Internet we know and use today.

2.4 BILLION OF THE 7 BILLION PEOPLE ON EARTH USE THE INTERNET

▶ 2010

3-D BODY PARTS

Invented in the USA, 3-D printing has been used since the 1980s to build up three-dimensional objects in layers from digital information. More recently, scientists have been developing 3-D printers to make human organs and body tissue.